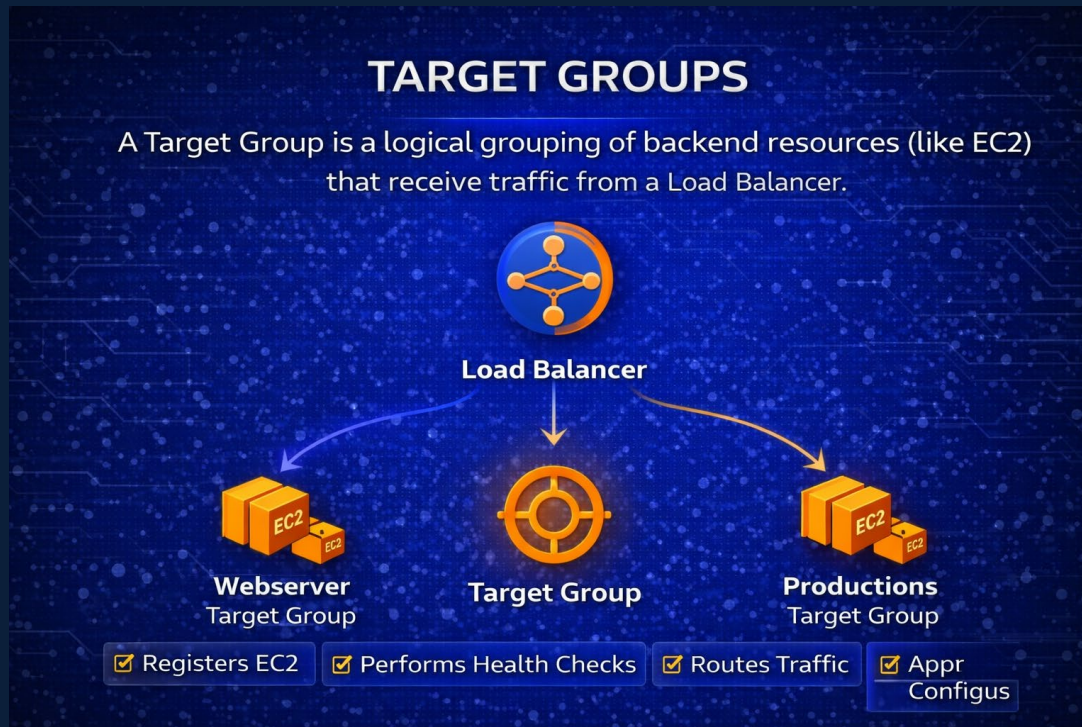


Traffic Routing & Health Management in Production Architectures

AWS EC2 Target Groups



Why Target Groups Matter



SINGLE EC2 IS NOT
SCALABLE



NEED TRAFFIC
DISTRIBUTION



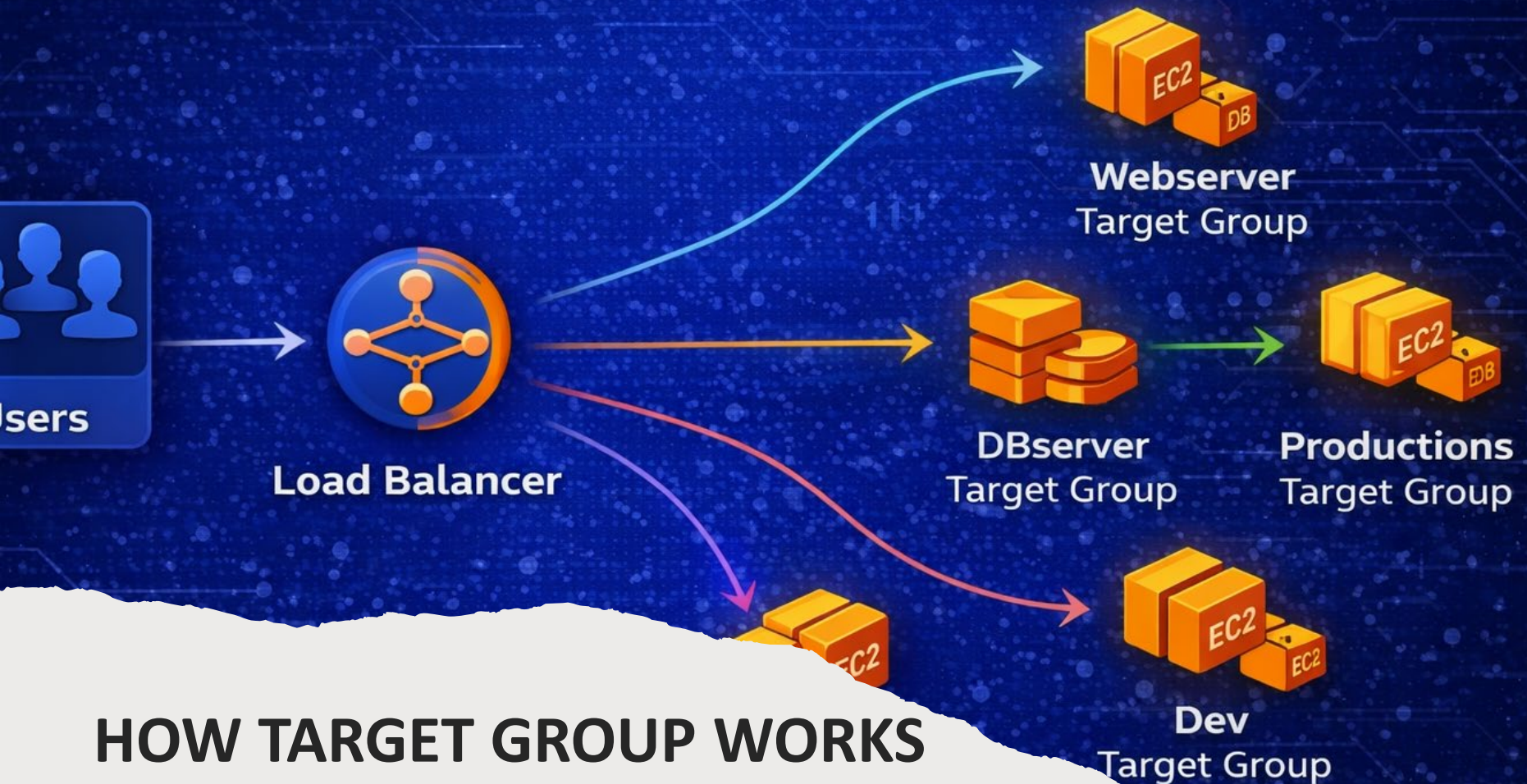
NEED AUTOMATIC
HEALTH
MONITORING



FOUNDATION FOR
HIGH AVAILABILITY

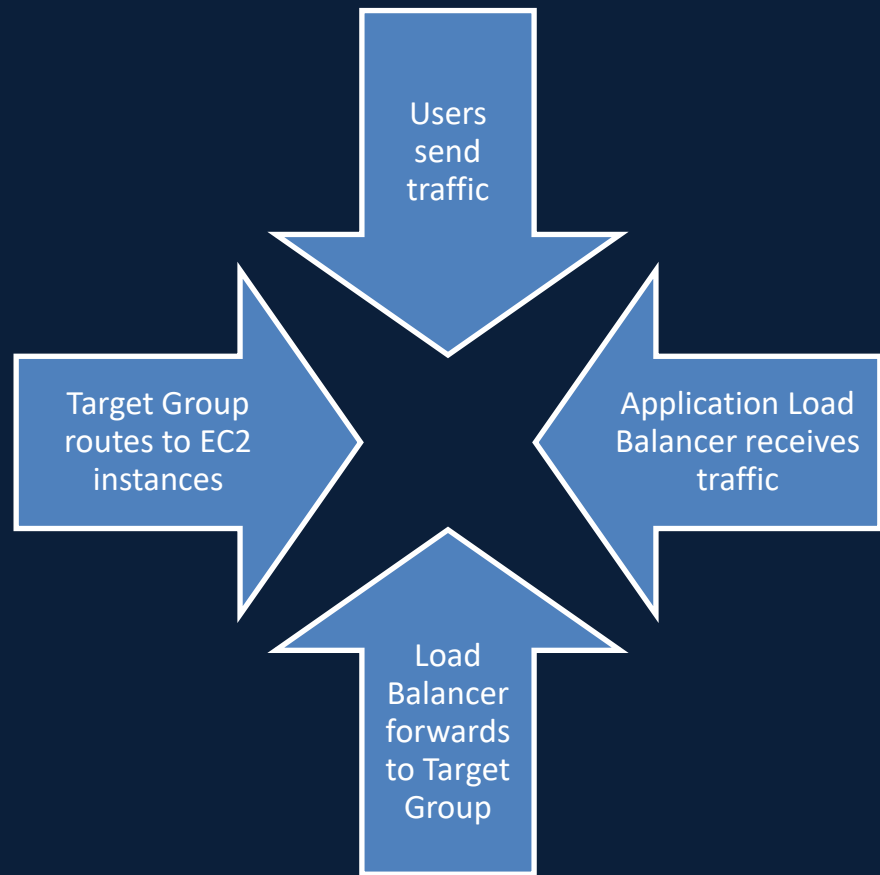
How Target Groups Work with Load Balancers

Routing Traffic to Different Groups of EC2 Targets



HOW TARGET GROUP WORKS

High-Level Architecture



What is a Target Group?

Logical grouping of backend resources

Registers EC2, IP, or Lambda

Performs health checks

Routes traffic from Load Balancer

Supported Target Types



EC2 INSTANCES
(MOST COMMON)



IP ADDRESSES

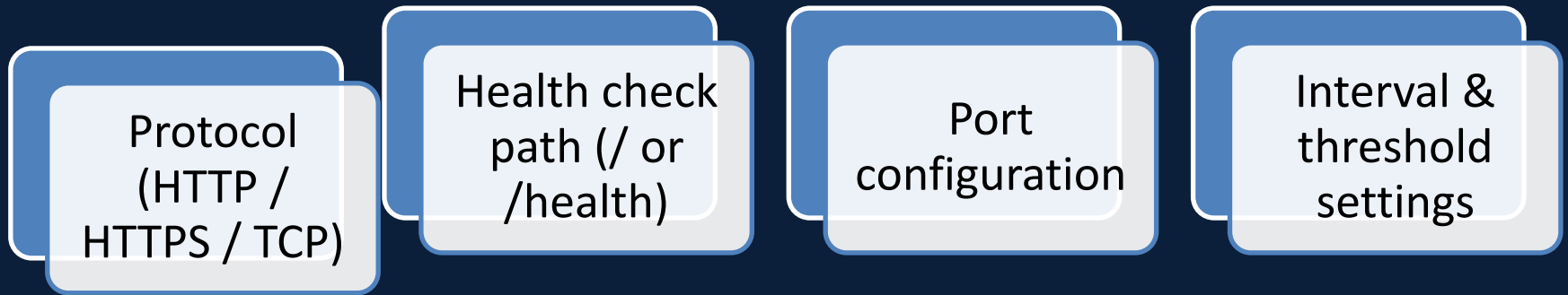


LAMBDA
FUNCTIONS



APPLICATION LOAD
BALANCER
(ADVANCED)

Health Checks (Critical Component)



How Health Checks Work



Traffic Routing Flow

Listener receives request
Listener rule evaluates path/host
Forwards request to target group
Target group selects healthy instance

Path-Based Routing Example

/api → TG-API

/images → TG-STATIC

/admin → TG-ADMIN

Supports microservices architecture

Target Group Attributes



Session stickiness



Deregistration delay



Slow start mode



Load balancing algorithm
(round robin)

Hands-On Lab Architecture

Create 2 EC2 instances

Install Nginx on both

Create Target Group (HTTP:80)

Register instances

Attach to Application Load Balancer

Create ALB (Internet-facing)

Create Listener (HTTP:80)

Attach Target Group

Test Load Balancer DNS

Failure Simulation

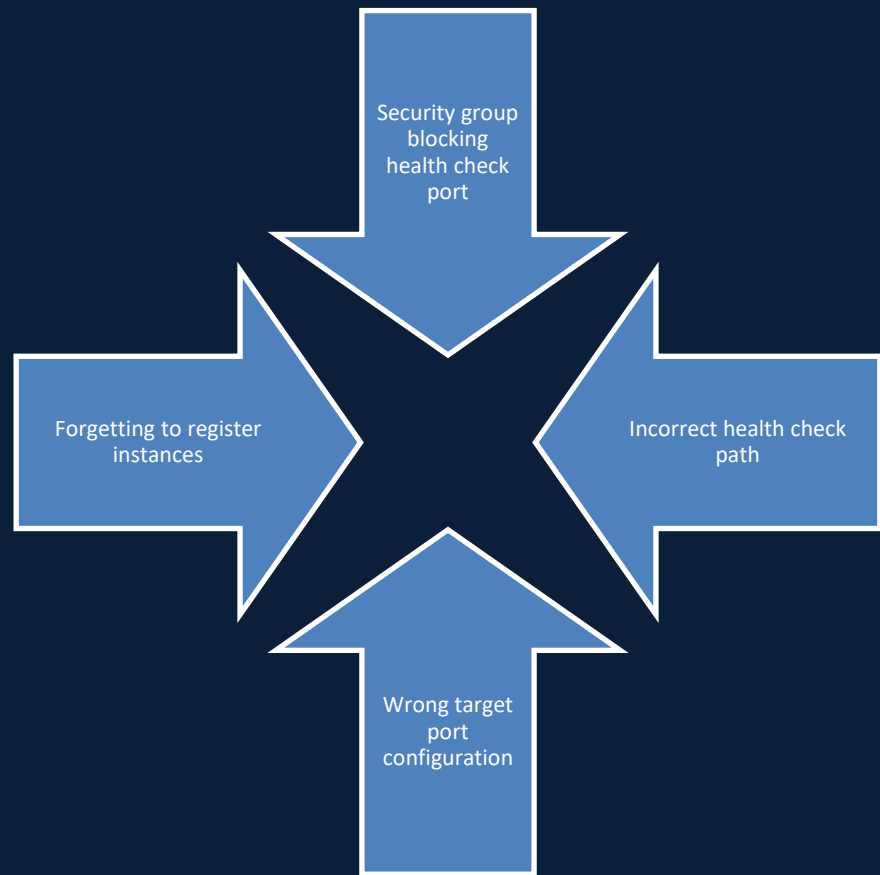
Stop one EC2 instance

Health check fails

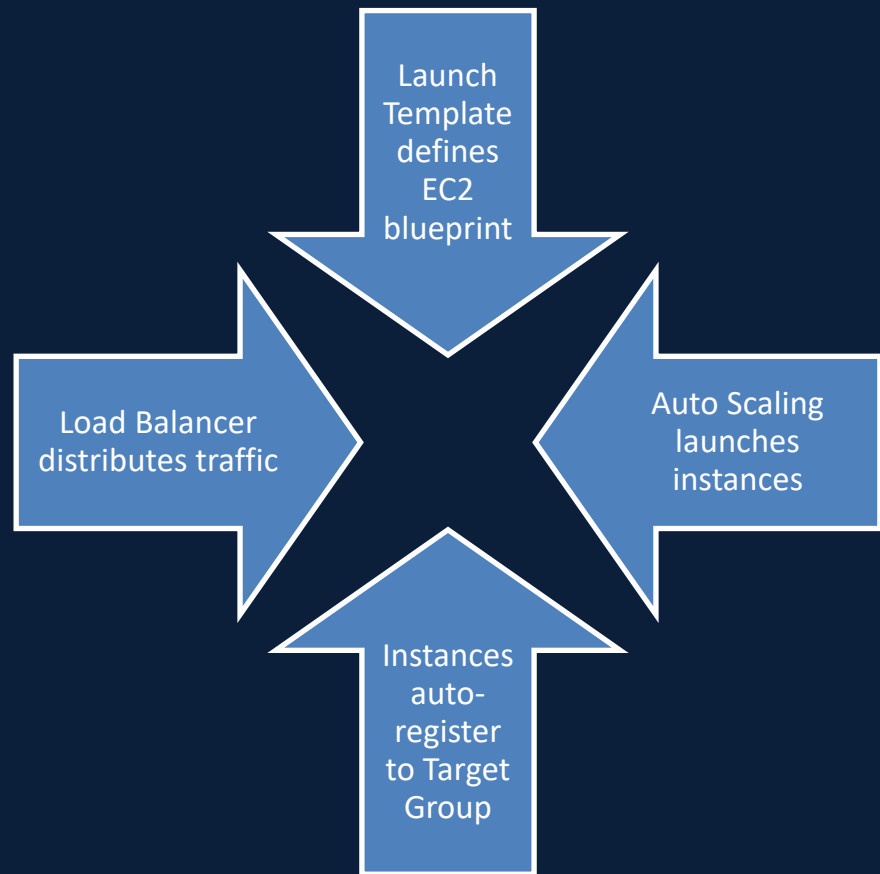
Traffic stops routing to failed instance

Application remains available

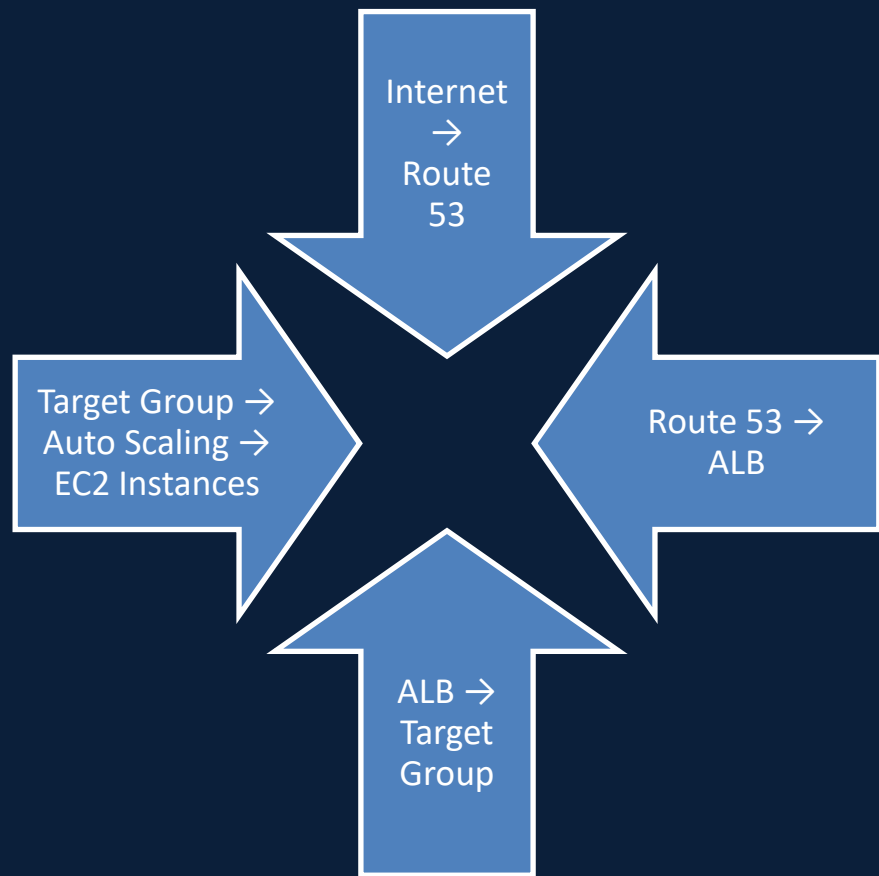
Common Configuration Mistakes



Target Groups + Auto Scaling



Production Architecture Overview



Session Summary

Target Groups manage backend traffic
Health checks ensure availability
Critical for scalable AWS systems
Essential knowledge for Cloud Engineers